

Open Science maturity for infrastructures

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Principle	Subprinciple	Maturity level 1	Maturity level 2	Maturity level 3	Maturity level 4
Openness	Results openly available	Data and content is not available.	Data and content is described with metadata. Data and content is findable.	Data and content is well described, findable and accessible with restrictions.	Data and content is well described, findable and accessible to everyone.
	Transparent processes	There is no documentation of what is happening or supposed to happen.	There is some guidelines on the process.	All steps in a process are described, recorded when executed and made available with restrictions.	All steps in a process are well described, recorded when executed and made publicly available.
	Open source	Only proprietary software is used.	Some core parts are using open source software.	Mostly open source software is used. Some open source software is contributed back to the community.	Full engagement in an open source community.
	Open to new ideas	At a stand still	Ad-hoc improvements	An innovation process is in place	Continuously innovating
	Open to participation				
Autonomy	Community governed	Decisions are made by a single party, without clear possibility for the community to participate.	Governance is shared by multiple parties, and there is an advisory board or similar to represent the community.	The governance is in the hands of a consortium or board that represents well the main stakeholders and user groups.	Full community participation in the governance through mechanisms like elections and self governance.
	Clear responsibilities				
	Reduced dependence on single supplier	The infrastructure uses mainly proprietary software.	The infrastructure provides an API for most functionalities.	It is possible to gradually migrate to another solution, e.g. adheres to open standards for integrations and data formats.	A full exit plan has been developed and there is a number of realistic alternatives.

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Sustainable	Financially secure	There is no funding at the moment and for the foreseeable future.	There is time-limited funding, e.g. through a one-off grant or donation.	There is for a number of years funding, e.g. through memberships and recurrent grants.	
	Human capacity available	No one is working on this infrastructure for a foreseeable time.	One hero is keeping the lights on, but there is no backup.	Somewhat staffed	Dedicated resourcing
	Legal and policy compliant	Unknown	Major gaps	Bases covered	Audited or certified
	Resilient	Falls over at the first breeze	Major gaps	Bases covered	Can weather a hurricane
	Institutionally embedded				
Researcher centric	Interconnected	This infrastructure has non integrations with others, and makes it difficult to even exchange data.	Uses other systems	Two-way connected	Part of flourishing ecosystem
	Seamless user-experience				
	Accessible to all				
	Recognition and rewards				